Fiscal Year:	FY 2015	**Task Last Updated:**	FY 11/04/2015
PI Name:	Frankfort, Benjamin J. M.D., Ph.D.		
Project Title:	ICP Measurement and Acute Modulation in Tree Shrews: A Feasibility Study for Long-Term ICP Elevation and Monitoring as a Model for Visual Impairment Intracranial Pressure (VIIP) Syndrome		
Division Name:	Human Research		
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Comments:			
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Norton, Thomas Ph.D. (University of Alabama at Birmingham) Twa, Michael O.D., Ph.D. (University of Alabama at Birmingham) Nusbaum, Derek M.D. (Baylor College of Medicine)		
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Task Description:	Specific Aims 1. Adapt an established technique of intracranial pressure (ICP) elevation in rodents to the tree shrew. We have previously developed a rodent-based system for the elevation of ICP via the placement of both infusion and monitoring catheters in the subarachnoid space (SAS). This system is robust, and can be used to modulate and measure ICP for a prolonged period of time. To determine the feasibility of analogous system in tree shrews, we propose to duplicate the surgical components of this procedure. With this approach, we can determine anatomic variances and landmarks of consequence, and establish the patency of the SAS in tree shrews. We will use the Atlas of the Tree Shrew by Tigges and Shantha to help identify the most appropriate location for the catheters. 2. Determine the volume-pressure relationship for ICP change in the tree shrew. Once we have established the patency of the system and placed both catheters successfully, we will allow the system to equilibrate so that baseline ICP can be measured. After this is achieved, we will change the volume of the SAS via the infusion and removal of known volumes of artificial cerebrospinal fluid (CSF), while simultaneously monitoring ICP directly. With this approach we can determine the volume-pressure relationship of the system and determine the volume and rate of artificial CSF infusion or removal needed to maintain ICP at a specific level. These experiments will inform future studies that will include the chronic infusion of artificial CSF through indwelling pumps. 3. Determine intraocular pressure (IOP) changes with acute ICP elevation. Once the ICP infusion and monitoring system has equilibrated, we will measure baseline IOP and continuously monitor IOP throughout the experiment using a separate pressure transducer connected to the anterior chamber of the eye. This will enable us to determine if acute changes in ICP impact IOP directly. We will also monitor IOP non-invasively in the fellow eye to determine if these monitoring methods are sensitive enough to observe any induced changes in IOP. Key Findings: • It is possible record ICP in a tree shrew • It is possible to measure capacitance in a tree shrew • We observed small, rapidly equilibrating changes in IOP with elevation of ICP • There is no evidence of communication between hemispheres • If tethering can be avoided, telemetry is likely possible Impact: Tree shrews are a potential analog for Vision Impairment and Intracranial Pressure (VIIP) syndrome, but there are considerable obstacles that will need to be overcome in order to establish the model. Proposed Plans: 1. Measure physiologic ICP a. Two stage experiment – implantation of catheter system followed by recovery b. After recovery record flow and ICP in a restraining system already in use in Dr. Norton's lab i. Awake restrained animal most likely to provide meaningful data at this stage 2. Determine volume flow relationship of ICP in an awake, restrained animal a. Restraining system as above b. Controlled infusion of known volumes of aCSF with real-time measurements of ICP c. Infusion to a set point of ICP and calculate volume required 3. Determine optic nerve anatomic changes with ICP alteration a. Use Spectralis OCT to measure anterior deformation of the lamina cribrosa in anesthetized animals (already available at University of Alabama at Birmingham-UAB)
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	Terrestrial models of VIIP syndrome, especially in animals with an intact lamina cribrosa, can be used to inform common human diseases of pressure variation, including idiopathic intracranial hypertension and glaucoma, as well as traumatic brain disease.
Task Progress:	Specific Aims 1 and 3 are largely complete, but additional animals may be required to confirm our findings. Specific Aim 2 requires additional study.
Bibliography Type:	Description: (Last Updated:)